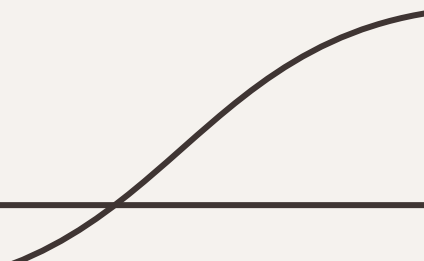
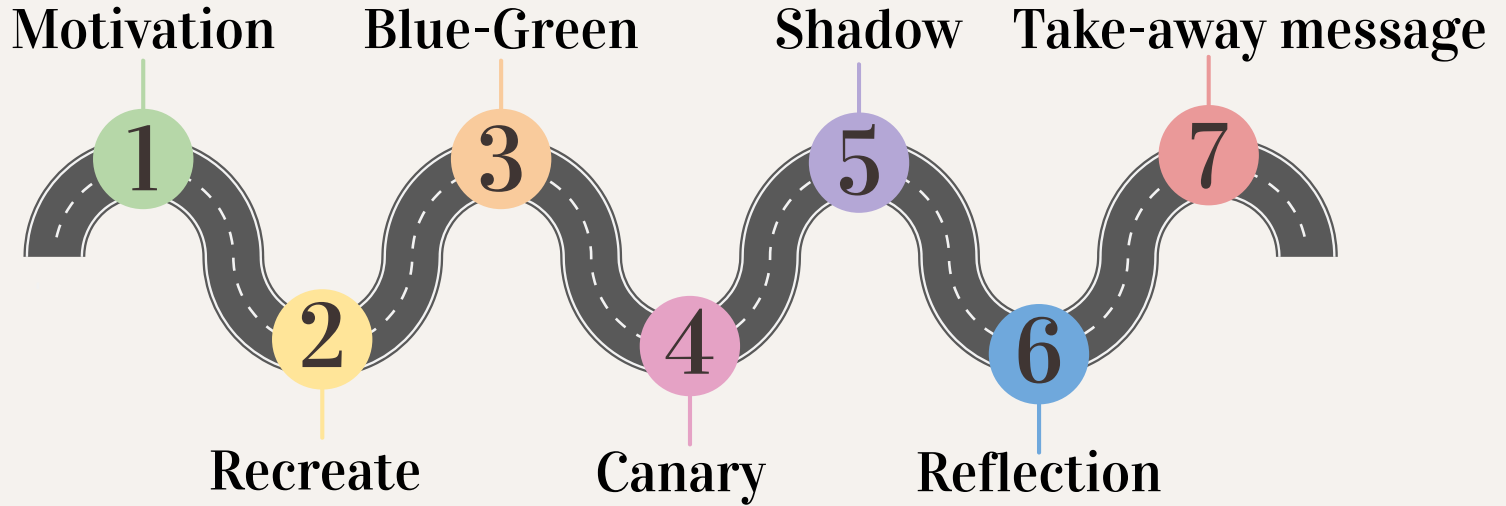


Comparison of Deployment strategies Shadow, Canary and Blue-Green

Edwin So
Þórfriður Ina Arinbjarnardóttir



Roadmap



Why Deployment Strategy matters



Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

**A LONG
TIME AGO
IN A GALAXY
FAR, FAR AWAY...**

imgflip.com

Motivation

Recreate

Blue-Green

Canary

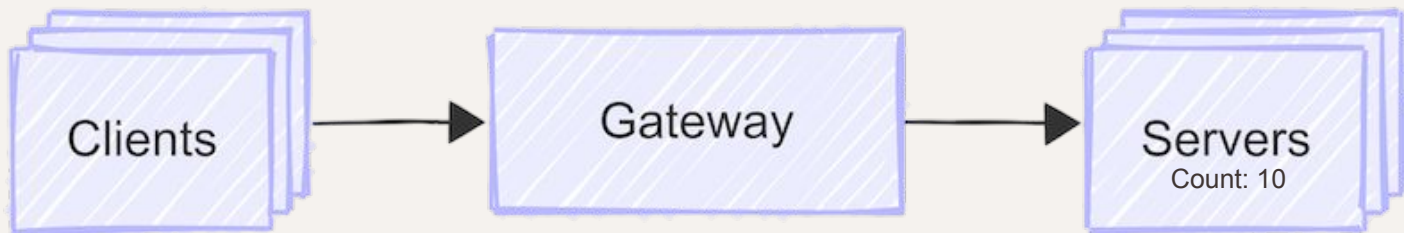
Shadow

Reflection

Take-away Message



System setup



```
http {  
    upstream backend {  
        server _default;  
    }  
  
    server {  
        location / {  
            proxy_pass http://backend;  
        }  
    }  
}
```

System setup



Motivation

Recreate

Blue-Green

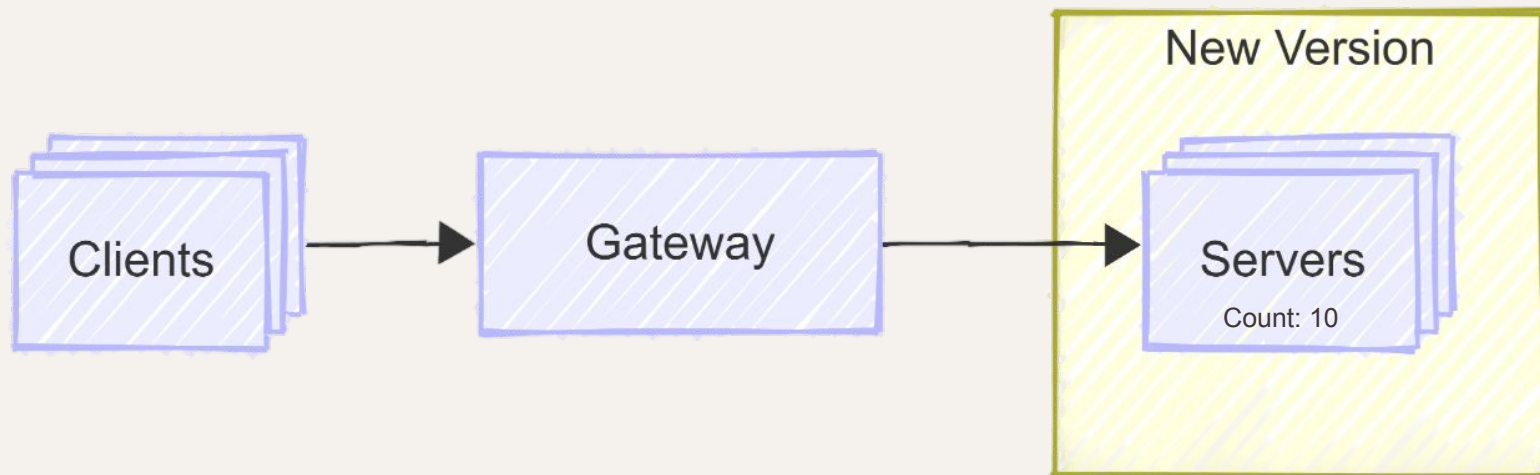
Canary

Shadow

Reflection

Take-away Message

Recreate



.... 1 hour later

Motivation

Recreate

Blue-Green

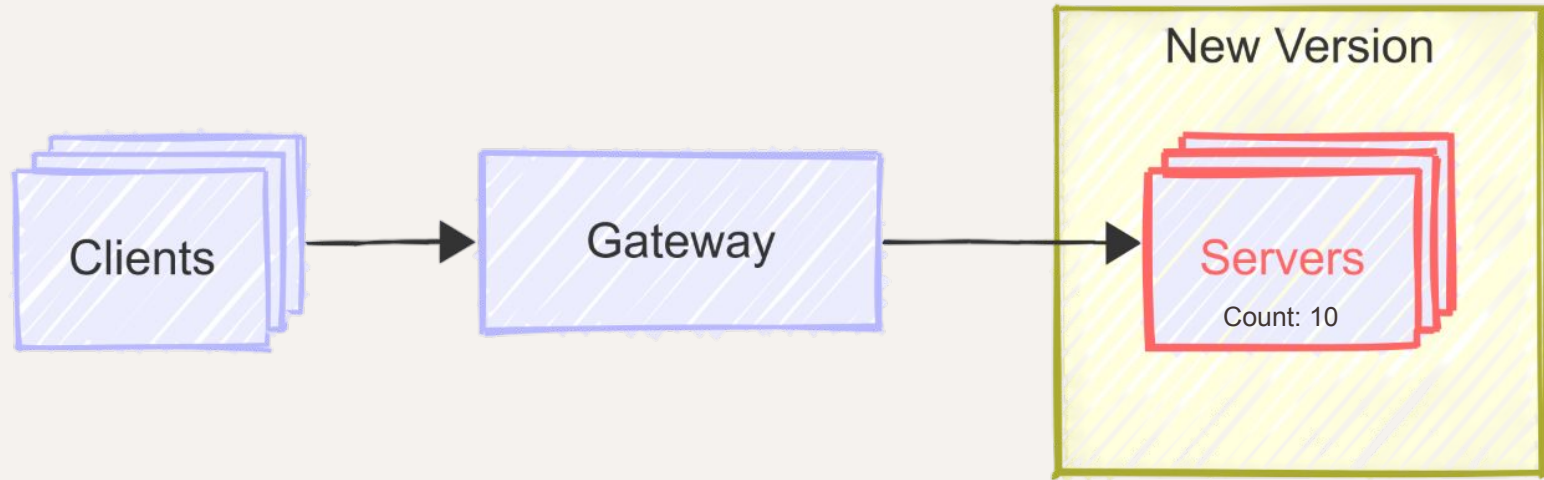
Canary

Shadow

Reflection

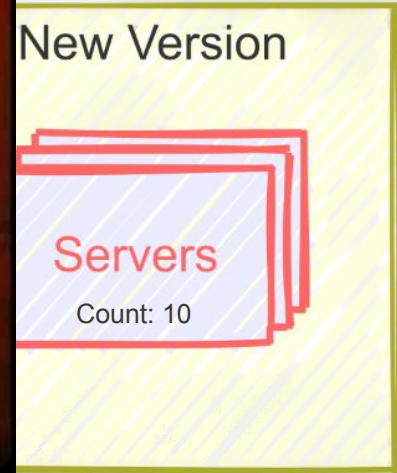
Take-away Message

Recreate



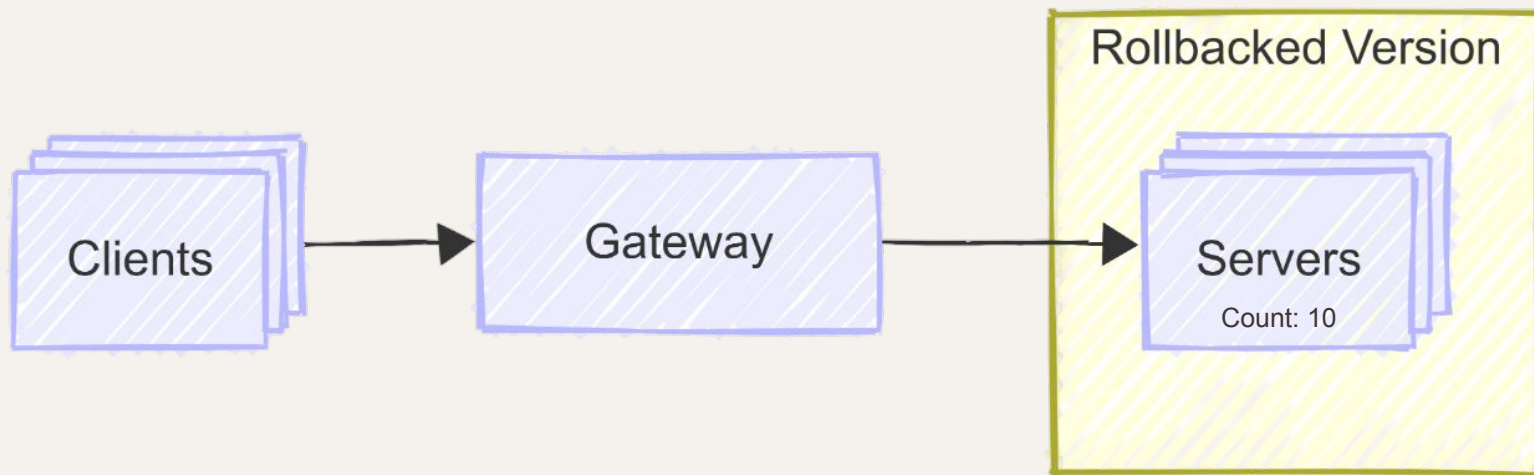
The new version has a bug.....

Recreate



The new version has a bug.....

Recreate



.... 1 hour later

Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

Aftermath



Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

Aftermath



*100% Availability is unrealistic - Checkout Google's SRE book, chapter Embracing Risk

Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

Aftermath



Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

HAVE YOU TRIED

BLUE/GREEN DEPLOYMENT?

imgflip.com

Motivation

Recreate

Blue-Green

Canary

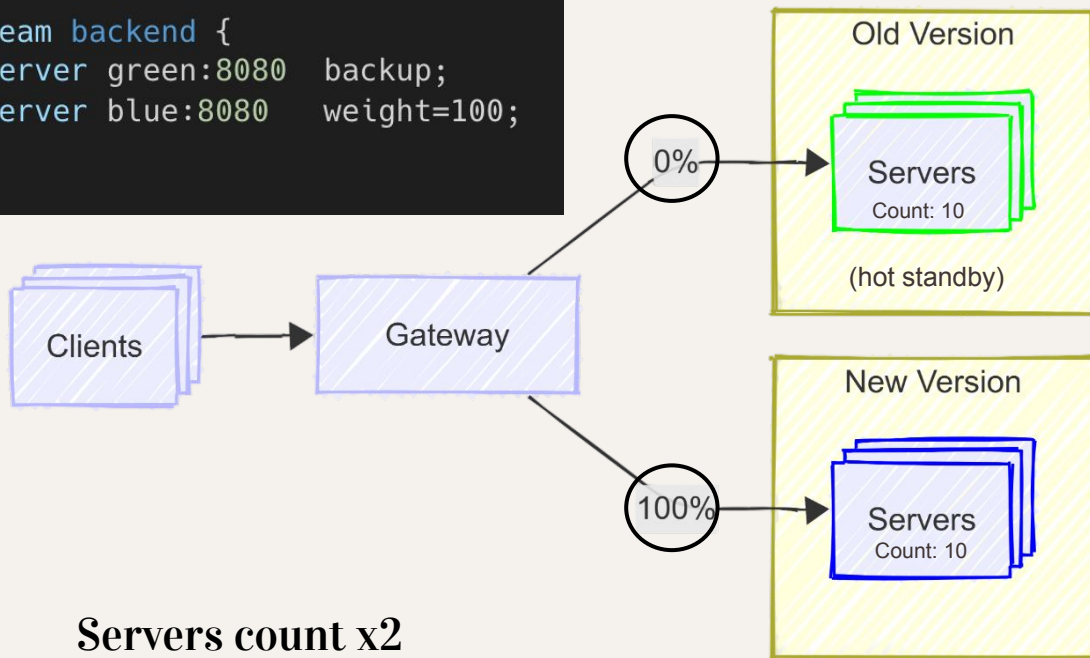
Shadow

Reflection

Take-away Message

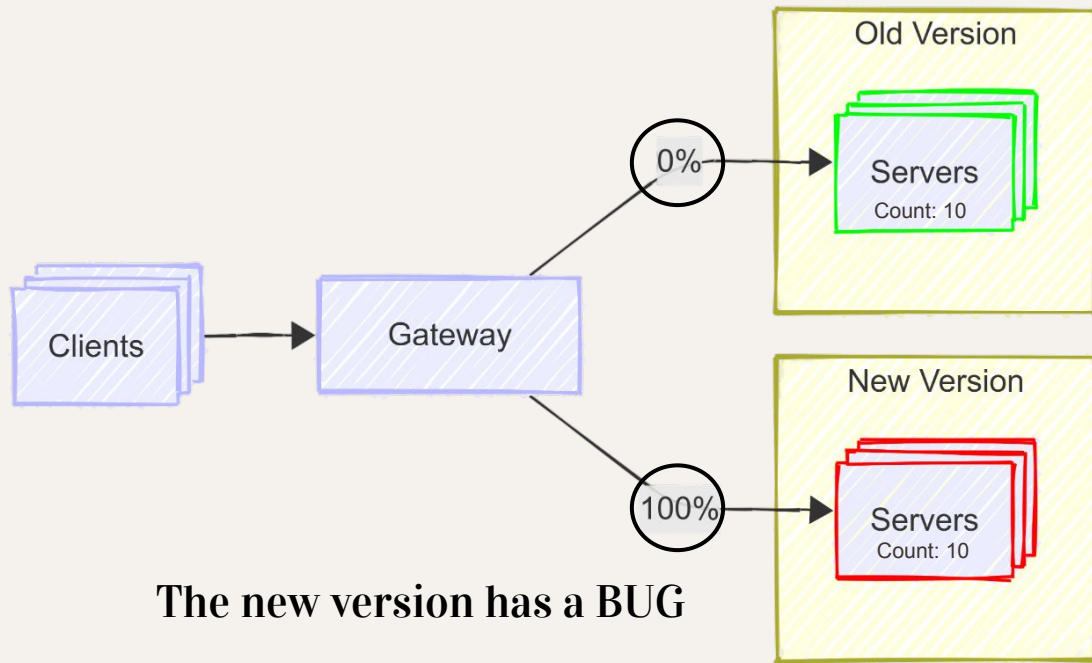
Blue-Green Deployment

```
upstream backend {  
    server green:8080 backup;  
    server blue:8080 weight=100;  
}
```



Servers count x2

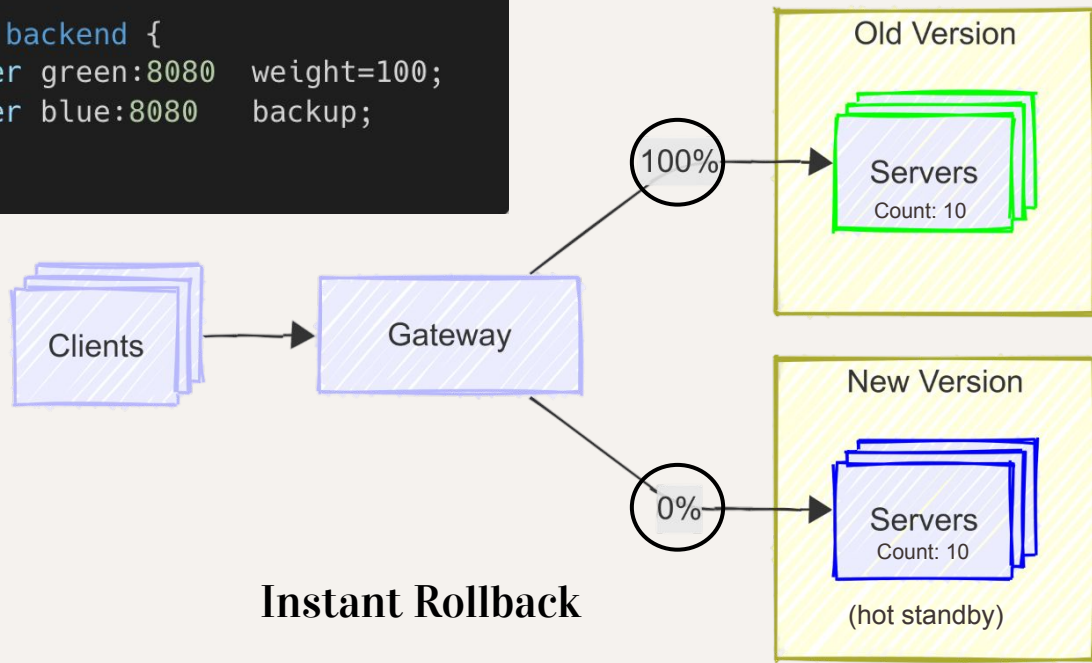
Blue-Green Deployment



The new version has a BUG

Blue-Green Deployment

```
upstream backend {  
    server green:8080 weight=100;  
    server blue:8080 backup;  
}
```



Instant Rollback

Blue-Green Deployment



Motivation

Recreate

Blue-Green

Canary

Shadow

Reflection

Take-away Message

Aftermath



Motivation

Recreate

Blue-Green

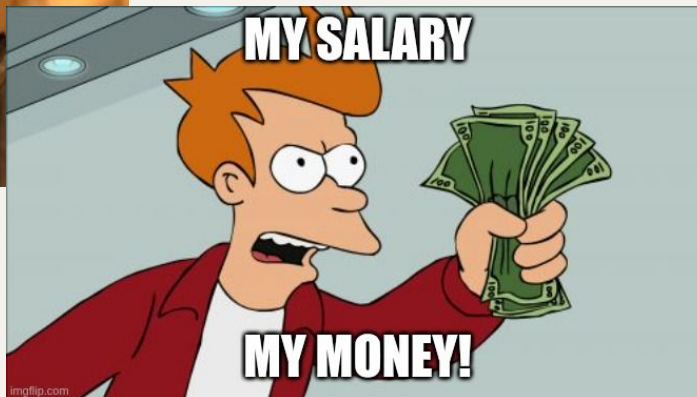
Canary

Shadow

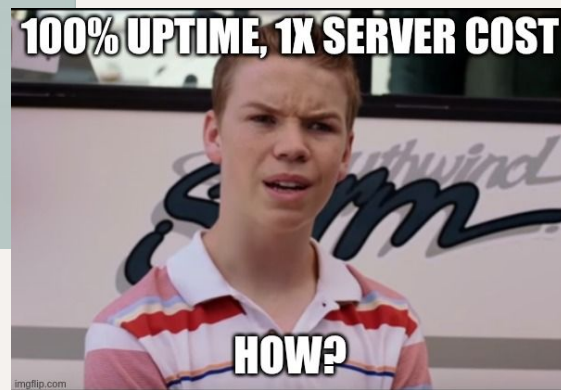
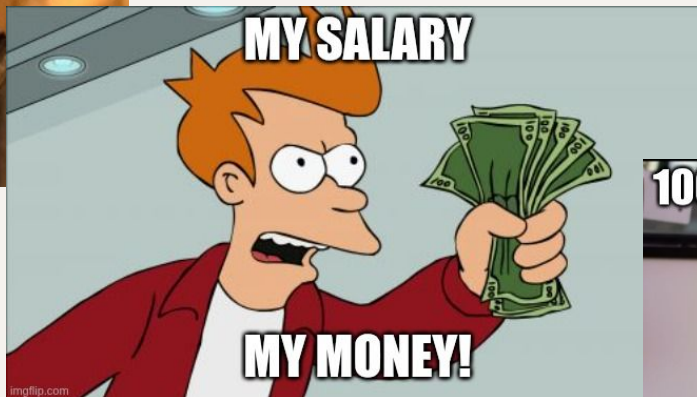
Reflection

Take-away Message

Aftermath



Aftermath



HAVE YOU TRIED

CANARY DEPLOYMENT?

imgflip.com

Motivation

Recreate

Blue-Green

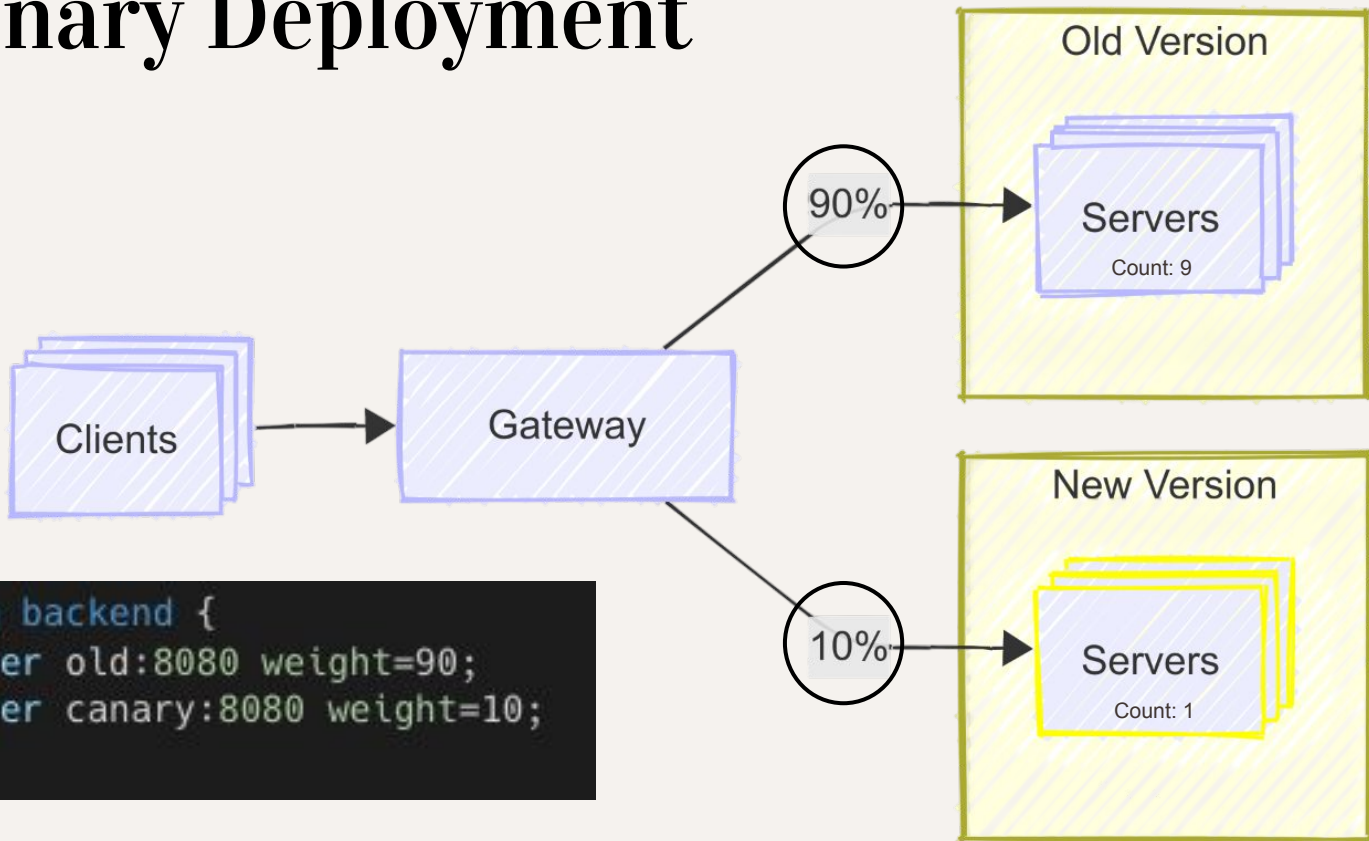
Canary

Shadow

Reflection

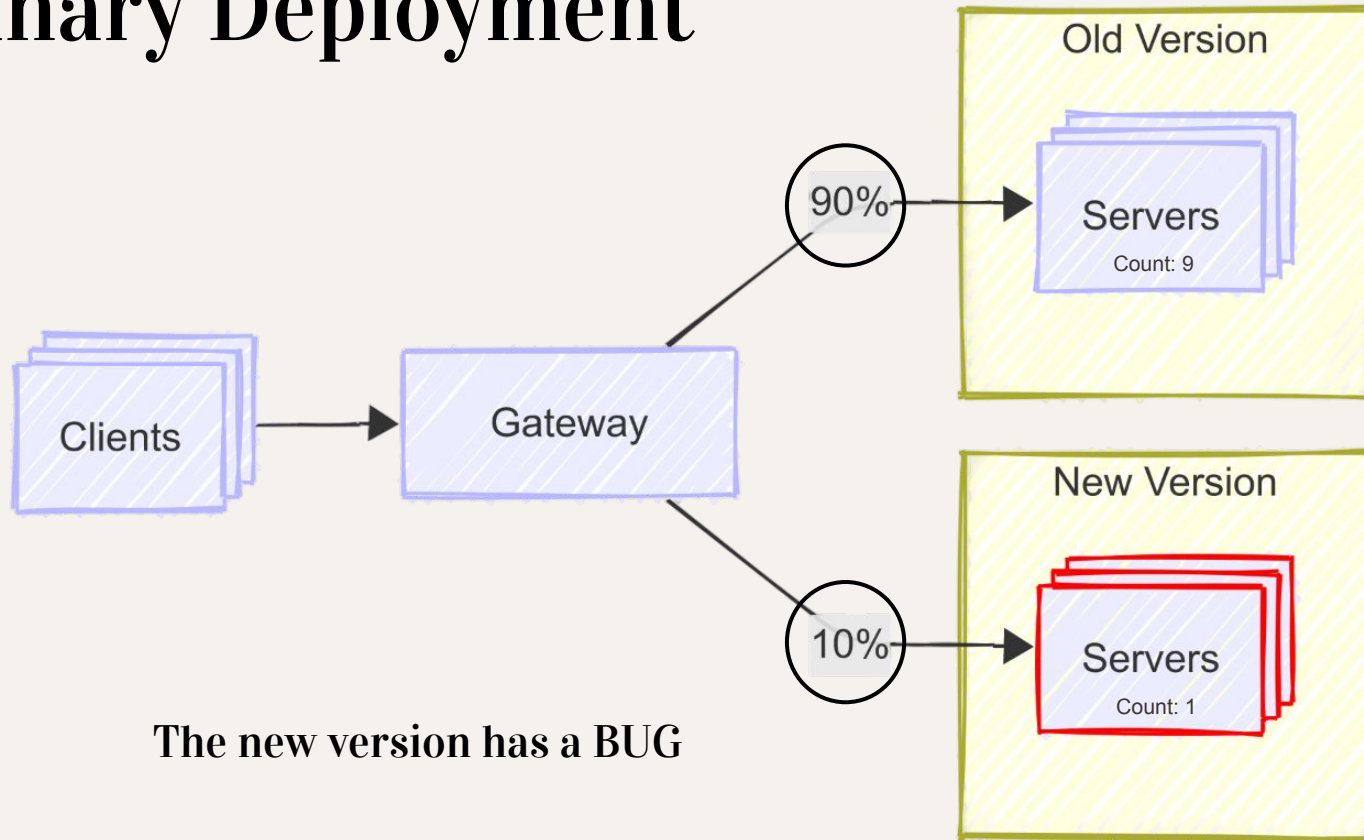
Take-away Message

Canary Deployment



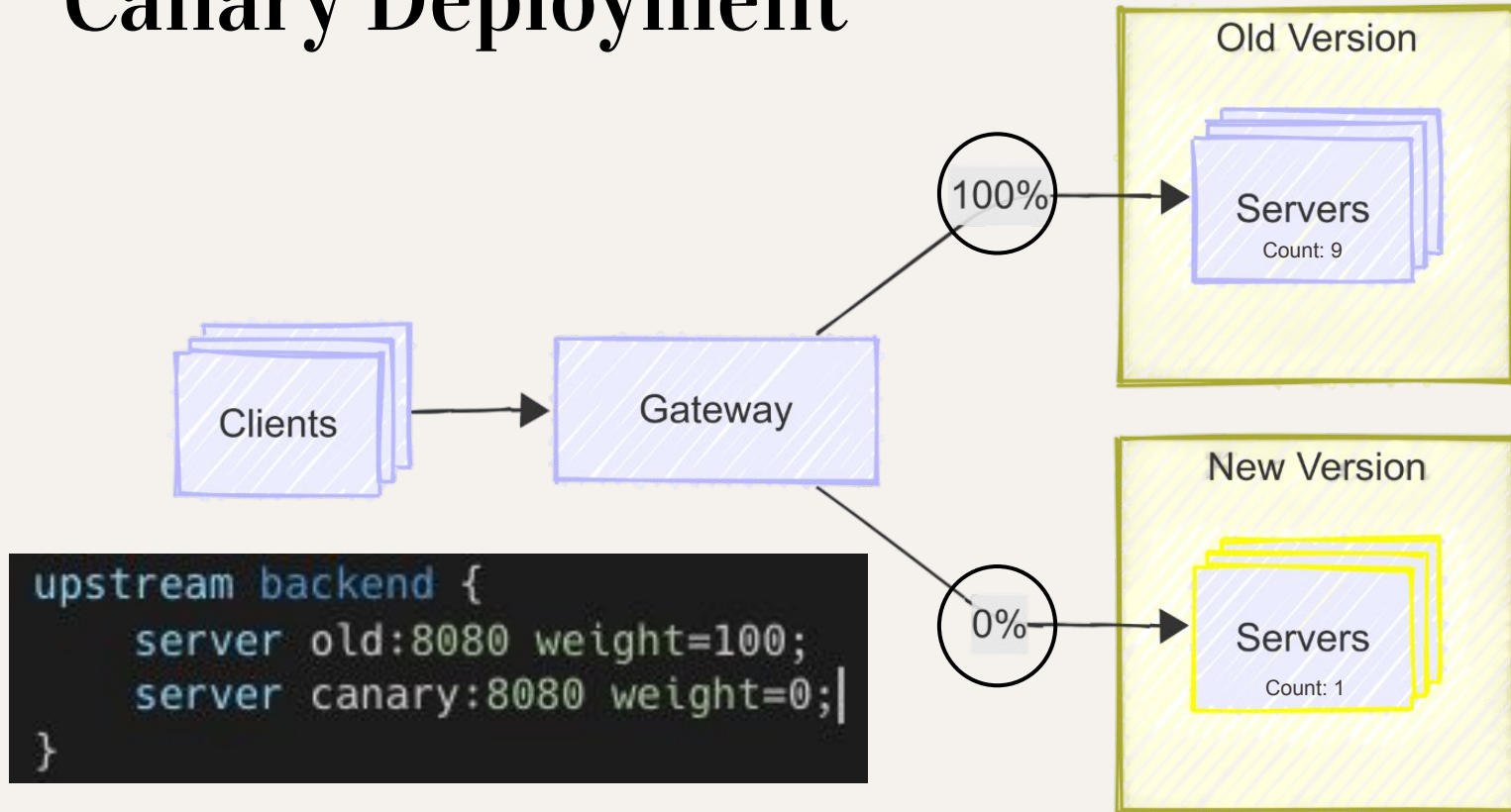
```
upstream backend {  
  server old:8080 weight=90;  
  server canary:8080 weight=10;  
}
```


Canary Deployment

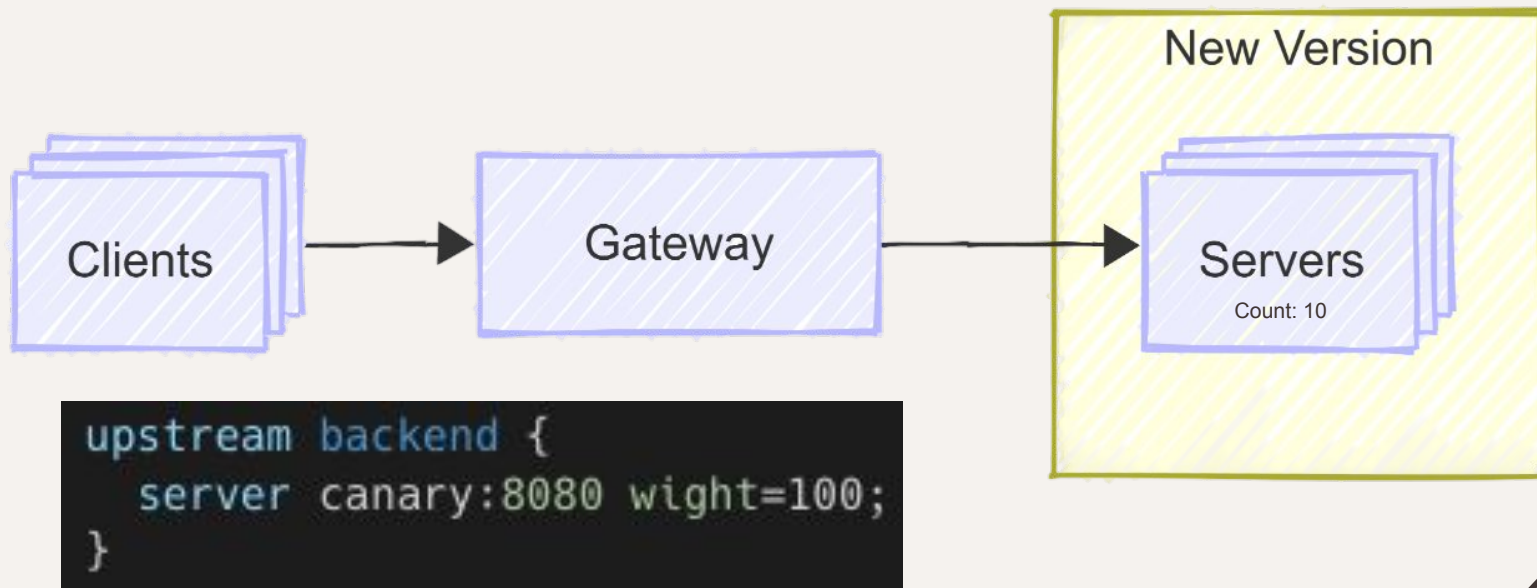


The new version has a BUG

Canary Deployment



Canary Deployment



Aftermath



Motivation

Recreate

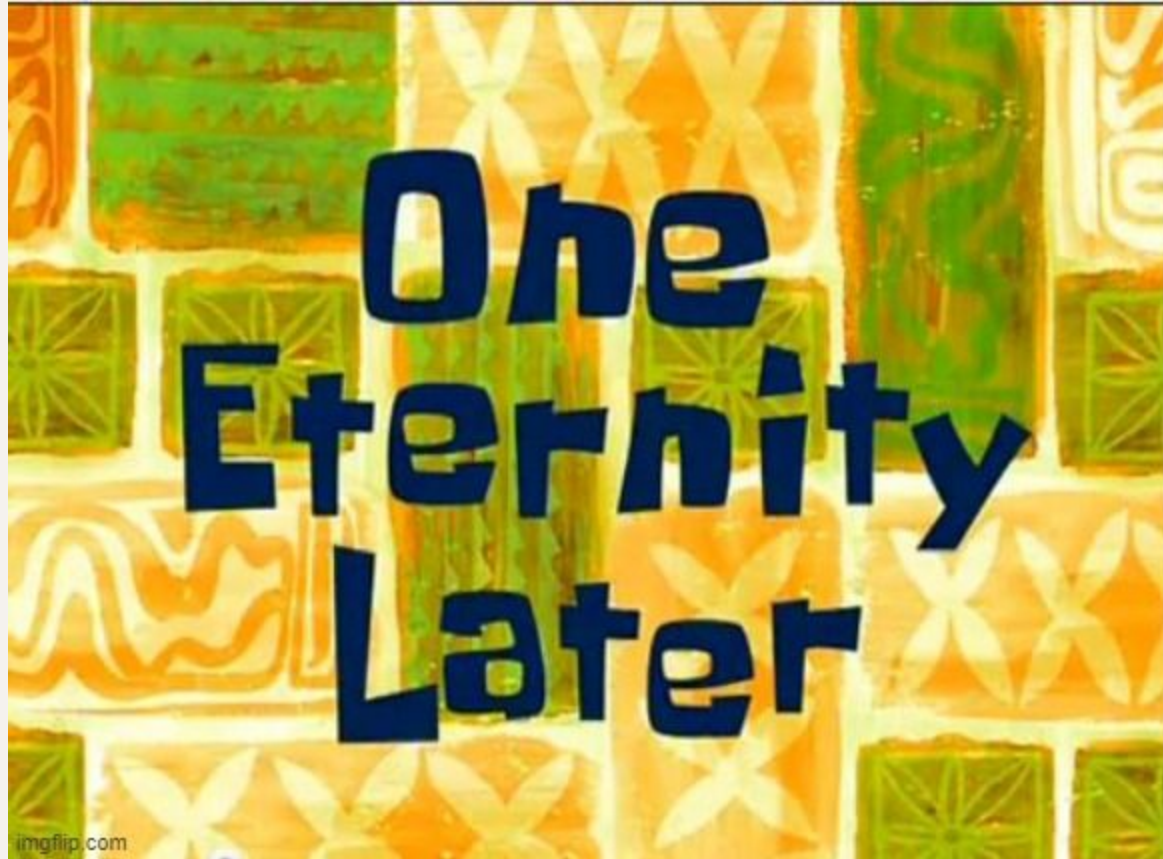
Blue-Green

Canary

Shadow

Reflection

Take-away Message





System Rewritten (JS $\rightarrow$ ASM)

Motivation

Recreate

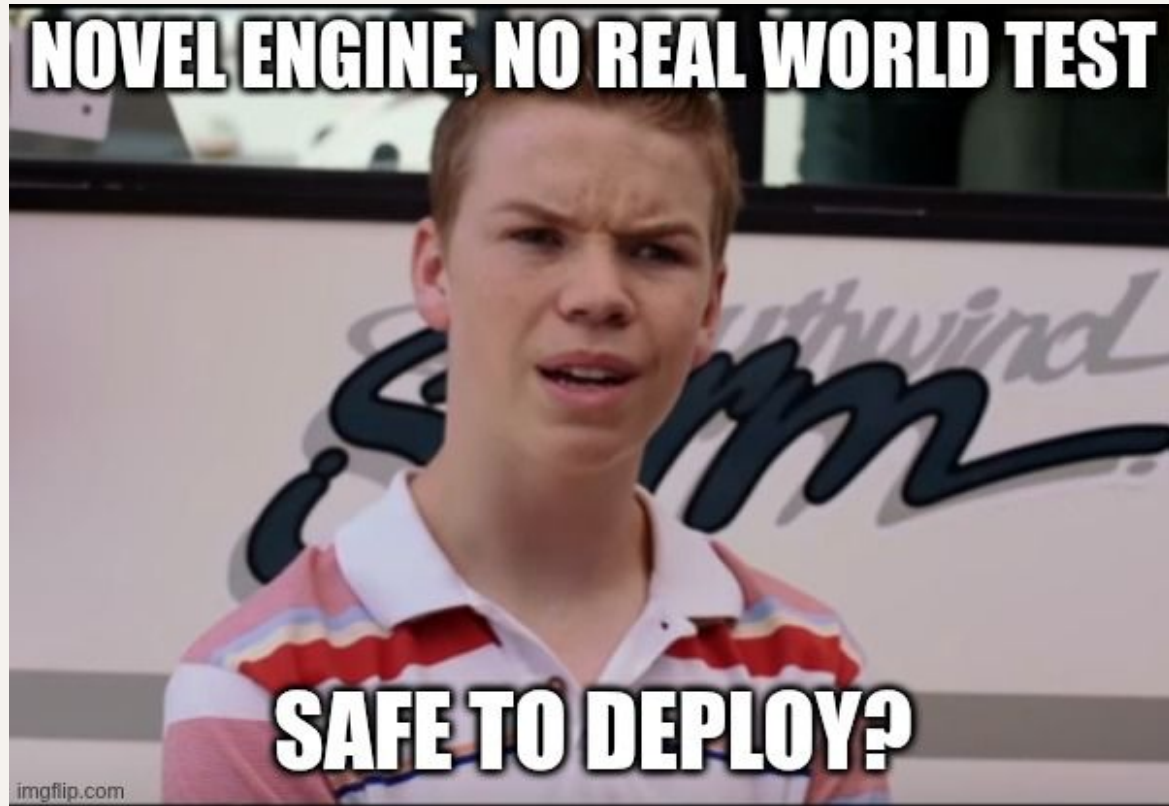
Blue-Green

Canary

Shadow

Reflection

Take-away Message



HAVE YOU TRIED

SHADOW DEPLOYMENT?

imgflip.com

Motivation

Recreate

Blue-Green

Canary

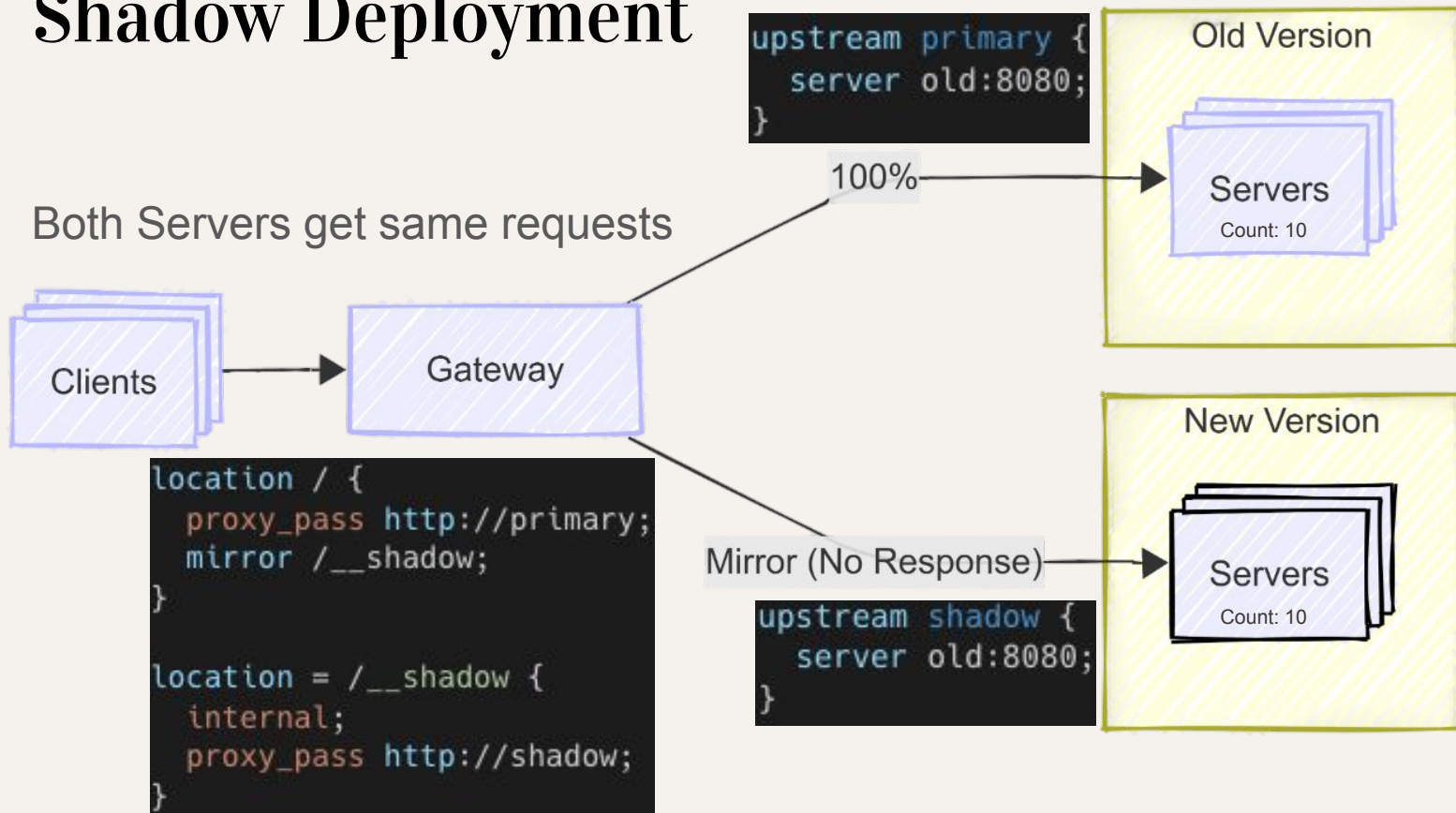
Shadow

Reflection

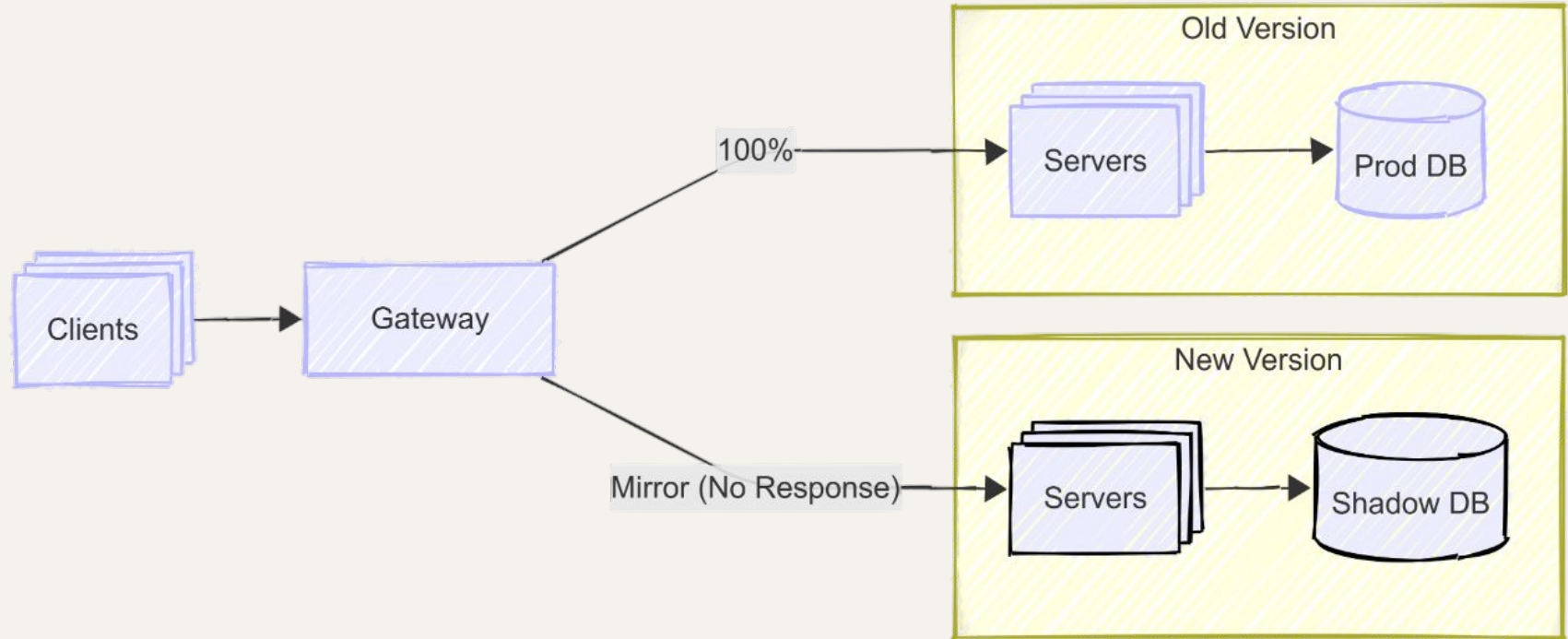
Take-away Message

Shadow Deployment

Both Servers get same requests



Shadow Deployment



Each version has its own state

Do all strategies share the same goal?



Comparison

	No downtime	Real traffic testing	Infracost	Negative User Impact	Complexity	Rollback duration
Recreate			\$			
Blue-Green			\$\$\$			
Canary			\$			
Shadow			\$\$\$			

Motivation

Recreate

Blue-Green

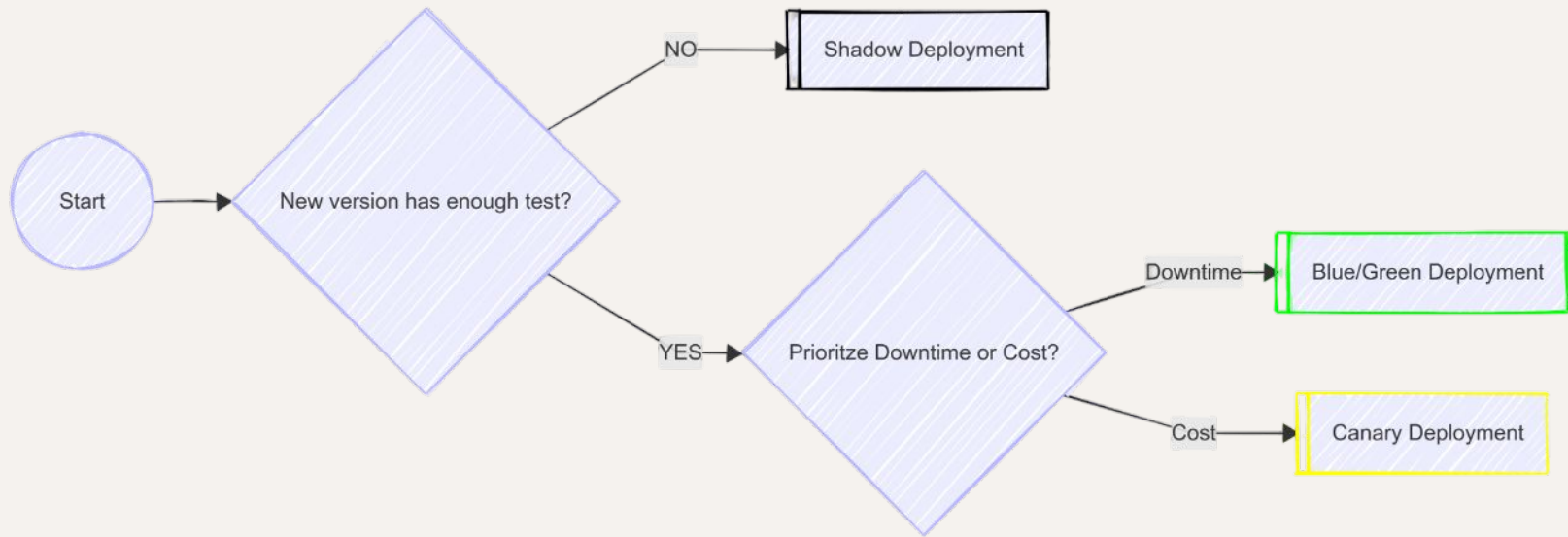
Canary

Shadow

Reflection

Take-away Message

How to choose a development strategy



Reflection



Take-away message

There is no single best deployment strategy, the right one depends on your situation, environment and priorities

